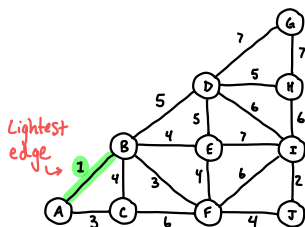


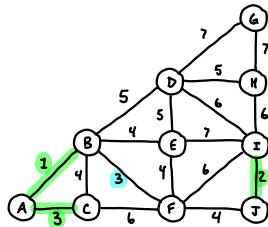
Kruskal's Algorithm

- Start with $T = \emptyset$
- Consider edges in increasing order of cost
- Insert edge e in T unless this would create a cycle.

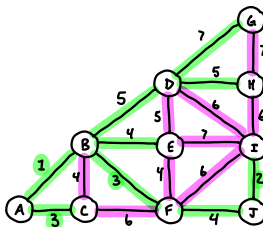
Every Node starts as a tree and whenever an edge it is connected to is chosen, it is absorbed into one larger tree. Eventually we have one final large tree that is the MST.



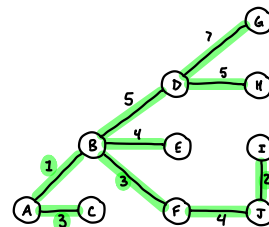
Lightest edge



Both BF and AC are new lightest

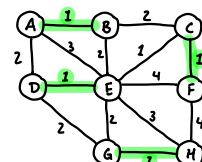
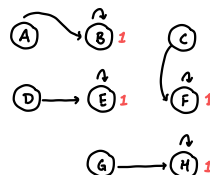
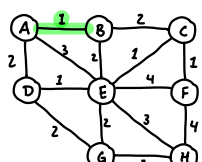
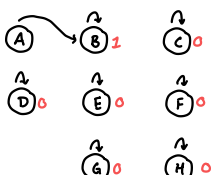
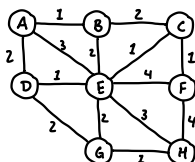
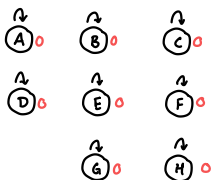


BC would make a cycle, so REMOVE it



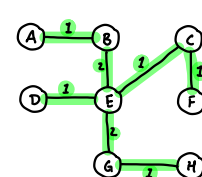
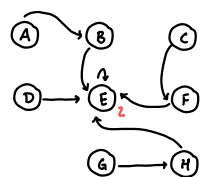
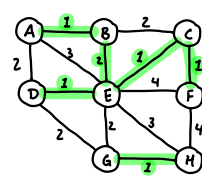
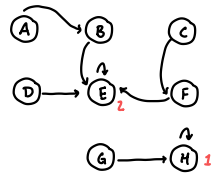
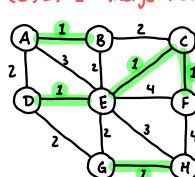
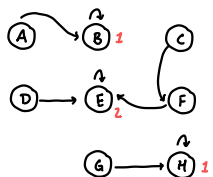
DSU Kruskal Example:

$\text{find}(A) \neq \text{find}(B)$
ranks are equal, so $\pi(A)=B, \text{rank}(B)+1$



Makeset each node

$(C, E) \in E$ (union)
Merge trees



root_u is F
Ranks are equal
root_v is E

F's new parent is E, since
the ranks were equal, $\text{Rank}(E)+1$.

Same as last step, but we DON'T
increase the rank of E, bc the ranks $\neq$.

